

CHIRAN BINNU CHERIAN

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EDUCATION

• Iowa State University, Ames, IA	May 2026 - Present
Ph.D. in Aerospace Engineering	GPA: 3.52/4.00
• Iowa State University, Ames, IA	Jan. 2024 - May 2026
M.S. in Aerospace Engineering, Thesis	GPA: 3.52/4.00
• Vellore Institute of Technology, Vellore, India	Jul. 2019 - May 2023
B.Tech. in Mechanical Engineering	GPA: 7.94/10.00

RESEARCH INTERESTS

Spacecraft Navigation & Estimation; Nonlinear and Bayesian Filtering; Uncertainty Quantification; Stochastic Optimal Control; Covariance Steering; Astrodynamics

RESEARCH EXPERIENCE

Spacecraft Procedures for Autonomous Control and Estimation Laboratory	May 2024 - May 2026
Graduate Researcher Iowa State University	M.S. Advisor: Dr. Simone Servadio

Nonlinear Estimation and Polynomial Kalman Filtering

- Co-developed **Quadratic Extended and Unscented Kalman Filters (QEKF/QUKF)** based on a quadratic approximation of the minimum mean square error estimator, extending conventional EKF and UKF measurement updates beyond an affine dependence on the measurement.
- Incorporated higher-order central moments, including **skewness and kurtosis**, into the measurement update to improve estimation under nonlinear and non-Gaussian uncertainty.
- Evaluated the proposed estimators on spacecraft relative-navigation problems using **Clohessy-Wiltshire dynamics** and non-Gaussian measurement noise, demonstrating improved accuracy over the corresponding linear-update filters.
- Demonstrated approximately **25% reduction in estimation error** for the QUKF relative to the UKF in the nonlinear scalar benchmark.

Higher-Order Polynomial Filtering and Conjugate Unscented Transformation

- Extended quadratic measurement updates to **arbitrary-order polynomial approximations of the Bayesian/MMSE estimator** while retaining the recursive structure of Unscented Kalman filtering.
- Integrated the **Conjugate Unscented Transformation (CUT)** to evaluate higher-order state and measurement moments using structured sigma-point sets.
- Developed and evaluated polynomial-CUT filtering architectures for **Clohessy-Wiltshire and Circular Restricted Three-Body Problem (CR3BP)** dynamics under non-Gaussian measurement uncertainty.
- Demonstrated improvements in both state-estimation accuracy and covariance consistency relative to conventional linear-update filtering methods.

Graduate Researcher Iowa State University	May 2026 - Present
Collaborators: Yasemin Isik	Ph.D. Advisor: Dr. Abhishek Halder

Risk-Sensitive Covariance Steering

- Formulated and analyzed the continuous-time **Linear Exponential Quadratic Gaussian (LEQG) covariance-steering problem** for stochastic linear systems with prescribed Gaussian endpoint covariances.

- Derived the structure of the optimal risk-sensitive state-feedback controller and its associated coupled matrix boundary-value formulation.
- Established **local existence and uniqueness** of the LEQG covariance-steering solution in a neighborhood of the corresponding risk-neutral solution.
- Developed a numerical solution procedure and demonstrated the framework on a noisy double-integrator system.
- Implemented a **Newton method with backtracking line search** for solving the implicit nonlinear matrix equation governing the optimal controller.

Composite Materials and Machine Learning

Undergraduate Researcher | Vellore Institute of Technology

May 2023 - May 2025

Collaborators: S. S. Kumar, S. V. S. Borkar, Y. Manikandan, et al. Advisors: Dr. Devasri Fuloria and Dr. Gaurav Gupta

- Investigated the mechanical and tribological characteristics of **sisal-fiber/dolomite-dust/epoxy hybrid composites** fabricated using hand lay-up techniques.
- Evaluated density, hardness, tensile strength, flexural strength, and sliding wear as functions of dolomite filler concentration.
- Applied **Taguchi L16 design of experiments** to characterize sliding-wear behavior and identify influential operating parameters.
- Evaluated **linear regression, decision tree, random forest, support vector regression, and artificial neural network** models for prediction of composite material properties; the ANN model provided the strongest predictive performance in the reported study.

PUBLICATIONS

Peer-Reviewed Publications

- **S. Servadio and C. B. Cherian**, “Quadratic Extended and Unscented Kalman Filter Updates,” *International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)*, 2025.
- D. Fuloria, **C. B. Cherian**, S. S. Kumar, S. V. S. Borkar, Y. Manikandan, A. Giri, P. R. Pati, A. Purohit, and G. Gupta, “Machine learning based prediction on mechanical and wear characterization of sisal fiber reinforced dolomite dust-epoxy hybrid composites,” *Journal of Materials Research and Technology*, vol. 41, pp. 3107-3132, 2026.

Preprints / Manuscripts

- **C. B. Cherian**, Y. Isik, and A. Halder, “Linear Exponential Quadratic Gaussian Covariance Steering,” arXiv:2609.12463, 2026.
- **C. B. Cherian** and S. Servadio, “Polynomial Updates for the Unscented Kalman Filter,” arXiv:2603.20259, 2026.

SELECTED TECHNICAL PROJECTS

Multi-Revolution Lambert Solver and Interplanetary Trajectory Reconstruction

MATLAB

Aug. 2024 - Dec. 2024

- Developed a constrained algorithm for solving **multiple-revolution Lambert problems**, incorporating periapsis and apoapsis restrictions for short- and long-path transfer solutions.
- Determined feasible semi-major-axis ranges satisfying orbital constraints for zero- and multiple-revolution transfer cases.
- Applied the framework to reconstruct the trajectory of the **Cassini-Huygens spacecraft** and validated generated spacecraft ephemerides against NASA/JPL Horizons data.
- Extended the trajectory framework to additional missions including Galileo and Voyager.

Titan Parafoil Terminal-Descent Optimization

Nonlinear Optimization

Feb. 2025 - May 2025

- Formulated a **nonconvex trajectory-optimization problem** for terminal descent and soft landing of a parafoil in a Titan mission scenario.
- Implemented **glide-slope, pointing-angle, and virtual-control constraints** to enforce safe descent while satisfying terrain-relative-navigation requirements.
- Evaluated the guidance methodology across varying initial position and velocity conditions and studied the interaction between physical and optimization constraints governing successful landing.

Continuum Damage Mechanics of Ductile Structures

ABAQUS

May 2024 - Aug. 2024

- Modeled ductile specimens in **ABAQUS** under plane-stress and plane-strain assumptions to investigate continuum damage mechanics and crack formation.
- Analyzed von Mises stress distributions, stress-strain behavior, and predicted crack evolution, comparing simulation behavior with published experimental and industry results.

Additive-Manufacturing Materials Study

PBF / DED

Mar. 2024 - May 2024

Conducted a comparative study of **AlSi10Mg** and **Ti6Al4V** manufactured using powder-bed fusion (PBF) and directed-energy deposition (DED), examining fatigue strength, yield behavior, fracture characteristics, and microhardness.

TEACHING EXPERIENCE

Iowa State University

Teaching Assistant for Linear Systems (AERE/EE/ME/MATH 5770)

Aug. 2025 - Dec. 2025

- Assisted with lectures, course presentations, assignment and examination grading, and student support for a graduate-level linear systems course.
- Conducted office hours and provided individual and small-group assistance on course concepts and problem solving.
- Developed a computational example to improve visualization and understanding of theoretically intensive linear-systems concepts.

INDUSTRY EXPERIENCE

Dynatomic Technologies Limited — Oldland Aerospace

Engineering Intern | India

Aug. 2023 - Oct. 2023

- Supported the design and manufacture of aerospace components using **SolidWorks**, including flap-track and helicopter-fuselage components.
- Conducted **quality control and non-destructive testing (NDT)** and supported compliance activities associated with FAA, aerospace, and ISO manufacturing requirements.
- Performed CNC machining, component assembly, and fabrication of metallic and composite specimens for destructive testing and finite-element-analysis validation.
- Collaborated with production personnel to identify manufacturing defects and support corrective improvements.

APT Global Marine Services L.L.C.

Engineering Intern | UAE

Jun. 2022 - Aug. 2022

- Supported manufacturing and testing of small naval components to meet customer design requirements.

- Produced CAD models in **Autodesk Fusion 360** and supported product-development activities from design through manufacturing.
- Gained hands-on experience with welding, CNC machining, assembly, and quality inspection for shipbuilding and naval-repair applications.

TECHNICAL SKILLS

Programming & Scientific Computing: Python, MATLAB, Mathematica

Estimation & Filtering: Kalman Filtering, EKF, UKF, QEKF, QUKE, Sigma-Point Methods, Conjugate Unscented Transformation, Monte Carlo Analysis

Optimal Control & Optimization: Stochastic Optimal Control, Covariance Steering, Nonlinear Optimization, Trajectory Optimization

Astrodynamics & Navigation: Clohessy-Wiltshire Dynamics, CR3BP, Lambert Problem, Orbit Determination, Spacecraft Relative Navigation

Engineering & Simulation: ABAQUS, ANSYS Fluent, SolidWorks, Autodesk Fusion 360, OpenRocket

Computational Tools: NumPy, SciPy, Matplotlib, Google Colab, Jetstream2

Other: Git, Linux, LaTeX

LEADERSHIP & ENGINEERING ACTIVITIES

Team Sammard — University Rocket Team, VIT

Senior Team Member; Former Recovery Head

Jul. 2019 - May 2023

- Contributed to the design, fabrication, launch, and recovery of a 10,000-ft sounding rocket developed for the Spaceport America Cup.
- Supported development of a tethered camera payload designed for deployment at approximately 4,000 ft for the CanSat Competition.
- Led and supported recovery-system activities enabling safe rocket recovery and reuse for subsequent testing and experimentation.
- Developed safer and more cost-effective procedures for ejection-system and motor testing and introduced tools including OpenRocket for design and mission analysis.

CERTIFICATIONS

Overview of Space Science (START Program) -- ISRO, 2023

SolidWorks Mechanical Design Certification -- Dassault Systèmes, 2022

Autodesk Certified User: Fusion 360 -- Certiport, 2022